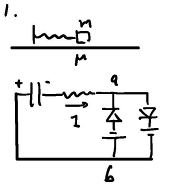
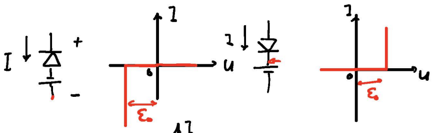
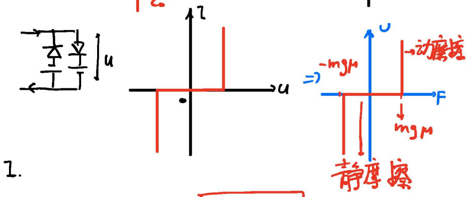
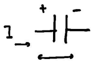
1.



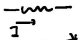






$\dot q = I.$

电容: $u=\frac{q}{C}$; $E=\frac{1}{2}\cdot\frac{1}{C}\cdot q^2 \sim \frac{1}{2}kx^2$; $F=kx$



电感 $u=L\cdot\frac{dI}{dt}$; $E=\frac{1}{2}\cdot L\dot I^2 \sim \frac{1}{2}m\dot x^2$ $F=m\ddot x$

$dW = u\cdot dq. \sim dW = F\,dx$

$\to \frac{1}{k}\cdot mg\mu\cdot 4$

$x=q=C\cdot\varepsilon_0\cdot 4.$

$k=\frac{1}{C}.$ $m=L.$ $\varepsilon_0=mg\mu.$ $\dot q>0$ → 简振. $U_C = q/C$

$L\ddot q = -\frac{1}{C}q-\varepsilon_0 \to q$ 简振. $\sim k\cdot x.$

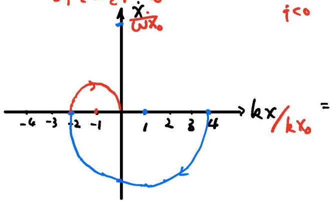
$$\begin{cases} m\ddot x = -kx - mg\mu & \dot x>0 \\ m\ddot x = -kx + mg\mu & \dot x<0 \end{cases}$$

$\to \ddot x = -\omega^2(x+x_0)$ $x_0=\frac{mg\mu}{k}.$

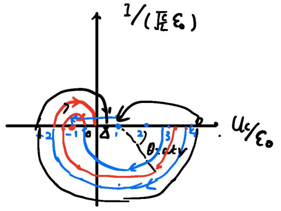
$\downarrow$

$\frac{1}{2}\dot x^2+\frac{1}{2}\omega^2(x+x_0)^2 = C.$

$L\ddot q = -\frac{1}{C}q+\varepsilon_0$ $\dot q<0$



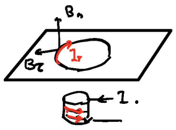
=)



Q 1.1
$U_0\in[0,\varepsilon_0)$ $U_1=U_0$
$U_0\in[\varepsilon_0,3\varepsilon_0)$ $U_1=2\varepsilon_0-U_0$
$U_0\in[3\varepsilon_0,4\varepsilon_0)$ $U_1=U_0-4\varepsilon_0$

之后 以$4\varepsilon_0$
为周期

2. 半径R. 电阻r. $L=0$



$\phi = MI$; $B_\tau = kI$

$I \quad 0 \to I_0$

$$I_r \cdot r = \mathcal{E} = \frac{d\phi}{dt} = M\frac{dI}{dt}.$$

$$F_\perp = I_r \cdot 2\pi R \cdot kI.$$

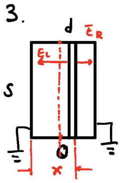
$$dp = F_\perp dt = \frac{M}{r}\frac{dI}{\cancel{dt}} \cdot 2\pi R \cdot kI \ \cancel{dt}$$

$$= \frac{M}{r} \cdot \pi R k\, dI^2$$

$$\Rightarrow p_{冲} = \frac{M}{r}\pi R k I_0^2 = mv_0$$

$$\Rightarrow h = \frac{v_0^2}{2g} = \frac{1}{2g}\cdot\left(\frac{M}{m}\frac{\pi R k I_0^2}{r}\right)^2$$ h小成立

3\.



$\varepsilon_r = 1+\chi$. $\quad \chi \propto n$.

p_0. T.

初态. $\varepsilon_{rL} = \varepsilon_{rR} = 1+\chi_0$.

动x. $\varepsilon_{rL} = 1+\chi_0\frac{d/2}{x}$

$\varepsilon_{rR} = 1+\chi_0\frac{d/2}{d-x}$.

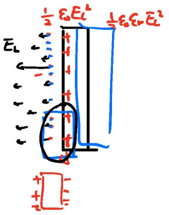
气体压强. 向右为正

$$\Sigma P_g = P_L - P_R = P_0\cdot\frac{d/2}{x} - P_0\frac{d/2}{d-x}$$

$$= P_0\, d/2 \cdot \frac{d-2x}{x(d-x)}. \qquad x>d/2.\ \Sigma P_g<0$$

①. $E_L \cdot x = E_R\cdot(d-x) = U_0$

自由电量Q. $D_{n_1} + D_{n_2} = Q/S$



$\frac{1}{2}\varepsilon_0 E_L^2$
$\frac{1}{2}\varepsilon_0\varepsilon_r E_L^2$
E_L

②. $\varepsilon_0\varepsilon_{rL}\cdot E_L + \varepsilon_0\varepsilon_{rR}\cdot E_R = Q/S$.

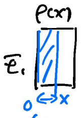
$$\frac{U_0}{x}\cdot\varepsilon_0\cdot\varepsilon_{r_L} + \frac{U_0}{d-x}\varepsilon_0\varepsilon_{rR} = Q/S$$

$$\Rightarrow U_0 = \frac{Q/S}{\frac{\varepsilon_0}{x}(1+\chi_0\cdot\frac{d/2}{x}) + \frac{\varepsilon_0}{d-x}(1+\chi_0\frac{d/2}{d-x})}$$

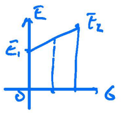
受力时表面极化电荷附加压给板. 固体 介质

$$\Sigma P_e = -\frac{1}{2}\varepsilon_0 E_L^2 + \frac{1}{2}\varepsilon_0 E_R^2.$$

$$\frac{1}{2}\varepsilon_0\varepsilon_r E^2$$
$$\uparrow$$
$$\frac{\sigma^2}{2\varepsilon_0\varepsilon_r}$$



$$\sigma(x) = \int_0^x \rho(\xi)d\xi$$



$\int E(x)\, d\sigma(x) \to p$.

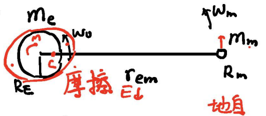
$\sigma_{总}\cdot\frac{E_1+E_2}{2} \to p$.

σ→ 自由电荷

$\sigma = D$.
$= \varepsilon_0\varepsilon_r E$

$$\Sigma P_e = \frac{1}{2}\varepsilon_0\cdot\left(\frac{1}{(d-x)^2} - \frac{1}{x^2}\right)\left[\frac{Q/S}{\frac{\varepsilon_0}{x}(1+\chi_0\cdot\frac{d/2}{x}) + \frac{\varepsilon_0}{d-x}(1+\chi_0\frac{d/2}{d-x})}\right]^2$$

4\.



$\omega_m = \frac{2\pi}{27.2}$

$\omega_m^2 = \frac{G(m_e+m_m)}{r_{em}^3}$ $\qquad G=$✓

地自 $\qquad$ 月自

$$L_{总} = \omega_0\cdot\frac{2}{5}m_e R_E^2 + \omega_m\cdot\frac{2}{5}m_m R_m^2$$

$$+ \frac{m_e m_m}{m_e+m_m}\cdot\omega_m\cdot r_{em}^2 \text{ 相对质心.}$$

$E_{总} = ?$

末态. $r_{em} \to \lambda r_{em}$.

$$\omega' \qquad \omega'^2 = \frac{G(m_e+m_m)}{(\lambda r_{em})^3} = \frac{1}{\lambda^3}\cdot\omega_m^2$$

$$\to \omega' = \omega_m \lambda^{-\frac{3}{2}} = x\,\omega_m$$

$x^2 = \frac{1}{\lambda^3}$.

$\lambda = x^{-\frac{2}{3}}$

$$L' = \omega'\cdot\frac{2}{5}m_e R_E^2 + \omega'\cdot\frac{2}{5}m_m R_m^2$$

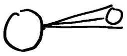
$$+ \frac{m_e m_m}{m_e+m_m}\omega'\cdot(r_{em}\lambda)^2$$

以$\omega_0 m_e R_E^2$为单位:

$$\frac{2}{5} + \frac{2}{5}\cdot\boxed{\frac{\omega_m}{\omega_0}\cdot\frac{m_m}{m_E}\cdot\left(\frac{R_m}{R_E}\right)^2} + \boxed{\frac{\omega_m}{\omega_0}\cdot\frac{m_m}{m_m+m_e}\cdot\left(\frac{r_{em}}{R_E}\right)^2}$$

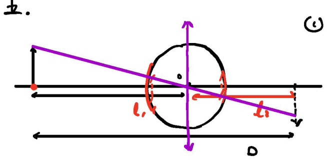
$$= \frac{2}{5}\cdot x + \frac{2}{5}x\boxed{\frac{\omega_m}{\omega_0}\cdot\frac{m_m}{m_E}\cdot\left(\frac{R_m}{R_E}\right)^2} + x\cdot\boxed{\frac{\omega_m}{\omega_0}\cdot\frac{m_m}{m_m+m_e}\cdot\left(\frac{r_{em}}{R_E}\right)^2}\cdot x^{-\frac{4}{3}}$$

$\omega' = \frac{2\pi}{52.46天}$. $\qquad r' = 1.545\, r_0$. $\qquad \lambda = 1.545$.



$$\frac{\Omega'}{\Omega} = \frac{(r-R_E)^2}{(r'-R_E)^2} = 0.41$$

五.



① l_1, l_2. $|l_1-l_2|=?$

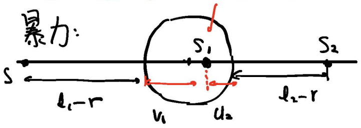
大小比值.

② $D >$ ____.

光路可逆. $l_1+l_2=D$.

$\frac{1}{l_1}+\frac{1}{l_2}=\frac{1}{f}$. 在哪无影响.

暴力:



$$\begin{cases}\frac{1}{l_1-r}+\frac{n}{v_1}=\frac{n-1}{r}\\ \frac{1}{l_2-r}+\frac{n}{u_2}=\frac{n-1}{r}\end{cases}\qquad \frac{n}{v_1}=\frac{n-1}{r}-\frac{1}{l_1-r}\qquad v_1=\frac{n}{\frac{n-1}{r}-\frac{1}{l_1-r}}.$$

$v_1+u_2=2r$

$$\frac{n}{\frac{n-1}{r}-\frac{1}{l_1-r}}+\frac{n}{\frac{n-1}{r}-\frac{1}{l_2-r}}=2r$$

$$\frac{n\cdot\cancel{r}(l_1-r)}{(n-1)(l_1-r)-r}+\frac{n\cancel{r}(l_2-r)}{(n-1)(l_2-r)-r}=2\cancel{r}$$

$$n(l_1-r)\cdot[(n-1)l_2-nr]+n(l_2-r)\cdot[(n-1)l_1-nr]$$
$$=2\cdot[(n-1)l_2-nr]\cdot[(n-1)l_1-nr].$$

$l_1\cdot l_2\ n\cdot(n-1)\cdot 2\quad -\underline{n^2r\cdot l_1}\ -l_2(n-1)rn$

$-2(n-1)^2 l_1 l_2\quad -n^2rl_2\ -l_1\underline{(n-1)rn}$

$+2\underline{nr(n-1)}l_1+2nr(n-1)l_2=0$

$l_1l_2\,2\cdot(n-1)\cdot$

$-l_1nr-l_2nr=0$

$\Rightarrow \frac{1}{l_1}+\frac{1}{l_2}=\frac{2(n-1)}{nr}=\frac{1}{f}$. → 等价于在O点单透镜.

↳ 等效焦距.

$$\begin{cases}l_1+l_2=D & ①\\ \frac{1}{l_1}+\frac{1}{l_2}=\frac{1}{f}\Rightarrow\frac{D}{l_1l_2}=\frac{1}{f} & l_1\cdot l_2=Df.\ ②\end{cases}$$

$①^2-4②\quad (l_1-l_2)^2=D^2-4Df$

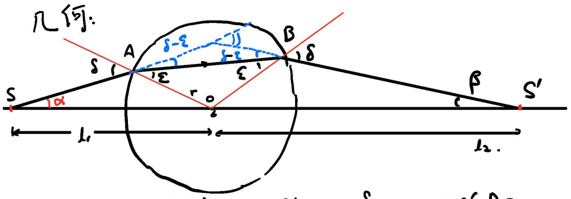
$\Rightarrow |l_1-l_2|=\sqrt{D(D-4f)}$. $D\geq 4f$. $f=\frac{nr}{2(n-1)}$

$|\frac{l_1}{l_2}|$. $x^2-Dx+Df=0$

$$x=\frac{D\pm\sqrt{D^2-4Df}}{2}$$

$$\left|\frac{l_1}{l_2}\right|=\left(\frac{D+\sqrt{D^2-4Df}}{D-\sqrt{D^2-4Df}}\right)^{\pm1}$$

几何:



△SAO. 正弦: $\frac{\alpha}{r}=\frac{\delta}{l_1}$ △S'BO, $\frac{\beta}{r}=\frac{\delta}{l_2}$.

$\Rightarrow$. $\delta=\frac{l_1}{r}\alpha$.

折射. $\delta=n\varepsilon$; $\varepsilon=\frac{l_1}{nr}\alpha$.

$\alpha+\beta=$ 偏转角 $=2(\delta-\varepsilon)$

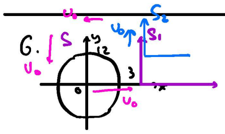
$=2\cdot(\frac{l_1}{r}\alpha-\frac{l_1}{nr}\alpha)=2\frac{l_1\alpha}{r}\ \frac{n-1}{n}$

$\alpha l_1=\beta l_2$

$\Rightarrow \frac{\cancel{\alpha}}{l_1}+\cancel{\alpha}\cdot\frac{\cancel{l_1}}{l_2}=2\cdot\frac{\cancel{l_1}\cancel{\alpha}}{r}\ \frac{n-1}{n}$

$\Rightarrow\frac{1}{l_1}+\frac{1}{l_2}=\frac{2(n-1)}{nr}$.

G.



(1) $S_2\to S_0$?

$x^0=ct$, $x^1=x$, $x^2=y$. $x^3=z$

x轴 $\begin{cases}ct=c\tau\\x=l\\y=0\\z=0\end{cases}$ $x^\mu(c\tau,l)$

在新系中 $x^{\mu\prime}(c\tau,l)$ 令 $x^{0\prime}=$ Const. $(x^1)'$. $(x^2)'$关系, x',y'

$S_0. \longrightarrow S_1 \longrightarrow S_2$

x轴 $\begin{cases}ct=c\tau\\x=l\\y=0\\z=0\end{cases}$ $\begin{cases}ct_1=\gamma(c\tau-\beta l)\\x_1=\gamma\cdot(l-\beta c\tau)\\y_1=0\\z_1=0\end{cases}$ $\begin{cases}ct_2=\gamma\cdot[\gamma(c\tau-\beta l)-\beta\cdot 0]\\x_2=\gamma(l-\beta c\tau)\\y_2=\gamma\cdot(0-\beta\cdot\gamma(c\tau-\beta l))\\z_2=0\end{cases}$

$ct_2=$ Const $=0$ 不妨为0

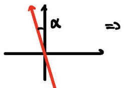
$c\tau=\beta l.\to\begin{cases}x_2=\gamma(l-\beta\cdot\beta l)=\frac{l}{\gamma} & \text{只相关于尺缩.}\\y_2=-\beta\gamma.(0-0)=0\end{cases}$

y轴. $S_0\to S_1\to S_2\to S_3$

$\begin{cases}ct=c\tau\\x=0\\y=l\\z=0\end{cases}$ $\begin{cases}ct=\gamma(c\tau-\beta\cdot 0)\\x=\gamma(0-\beta c\tau)\\y=l\\z=0\end{cases}$ $\begin{cases}ct=\gamma\cdot[\gamma c\tau-\beta l]\\x=-\gamma\beta c\tau\\y=\gamma(l-\beta\cdot\gamma c\tau)\\z=0\end{cases}$ $\begin{cases}ct=\gamma\cdot[\gamma(\gamma c\tau-\beta l)+\beta\cdot\gamma(l-\beta\gamma c\tau)]\\x=\gamma[-\gamma\beta c\tau+\beta\cdot\gamma(\gamma c\tau-\beta l)]\\y=\gamma(l-\beta\gamma c\tau)\\z=0\end{cases}$

S_2: $ct=0$; $\gamma c\tau=\beta l$.

$\begin{cases}x=-\gamma\beta\cdot\frac{\beta l}{\gamma}=-\beta^2 l\\y=\gamma(l-\beta\cdot\beta l)=\frac{l}{\gamma}\end{cases}$



$\Rightarrow \tan\alpha=\beta^2\cdot\gamma$

转角. $\downarrow S_4$

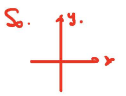
$\begin{cases}ct_4=\gamma[\gamma[\gamma(\gamma c\tau-\beta l)+\beta\gamma(l-\beta\gamma c\tau)]+\beta\gamma(l-\beta\gamma c\tau)]\\x_4=\gamma\cdot[-\gamma\beta c\tau+\beta\gamma(\gamma c\tau-\beta l)]\\y_4=\gamma[\gamma(l-\beta\gamma c\tau)+\beta\cdot\gamma[\gamma(\gamma c\tau-\beta l)+\beta\gamma(l-\beta\gamma c\tau)]]\\z_4=0\end{cases}$

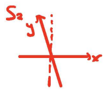
二阶: S_2: $\begin{cases}ct_2=(1+\beta^2)c\tau-\ \cdot\beta\cdot l\\x_2=-\beta\cdot\ \cdot c\tau\\y_2=(1+\frac{\beta^2}{2})l-\beta\cdot c\tau\end{cases}$

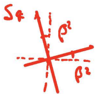
S_3: $\begin{cases}ct_3=\gamma[(1+\beta^2)c\tau-\ \cdot\beta\cdot l]+\beta\cdot(-\beta c\tau)\\ \quad=(1+\frac{1}{2}\beta^2)c\tau-\beta l\\x_3=\gamma[-\beta c\tau+\beta[(1+\beta^2)c\tau-\beta l]\\ \quad=O(3)c\tau-\beta^2 l\\y_3=(1+\frac{\beta^2}{2})l-\beta c\tau\end{cases}$

S_4: $\begin{cases}ct_4=\gamma\cdot[(1+\frac{\beta^2}{2})c\tau-\beta l+\beta\cdot[(1+\frac{\beta^2}{2}.l)-\beta c\tau]\\ \quad=c\tau.\\x_4=-\beta^2 l.\\y_4=\gamma[[(1+\frac{\beta^2}{2})l-\beta c\tau]+\beta([1+\frac{1}{2}\beta^2)c\tau-\beta l)]\\ \quad=O(3)c\tau+l.\end{cases}$

$\begin{cases}x_4=-\beta^2 l\\y_4=l\end{cases}$ → 转 $\alpha=\beta^2$ 角.







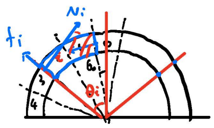
矩阵表达:

$$\begin{bmatrix} ct_1 \\ x_1 \\ y_1 \\ z_1\end{bmatrix} \begin{matrix}=\\=\\=\\=\end{matrix} \begin{bmatrix}\gamma & -\gamma\beta & +0 & +0\\ -\gamma\beta & +\gamma & +0 & +0\\ 0 & +0 & +1 & +0\\ 0 & +0 & +0 & +1\end{bmatrix}\cdot\begin{bmatrix}ct\\ \gamma\\ y\\ z\end{bmatrix}$$

$$\begin{bmatrix} ct_2 \\ x_1 \\ y_2 \\ z_2\end{bmatrix} = \begin{bmatrix}\gamma & 0 & -\beta\gamma & 0\\ 0 & 1 & 0 & 0\\ -\beta\gamma & 0 & \gamma & 0\\ 0 & 0 & 0 & 1\end{bmatrix}\cdot\begin{bmatrix}ct_1\\ x_1\\ y_1\\ z_1\end{bmatrix} = \begin{bmatrix} & \\ & \end{bmatrix}\cdot\begin{bmatrix} & \\ & \end{bmatrix}\cdot\begin{bmatrix}ct\\ x\\ y\\ z\end{bmatrix}$$

$$\begin{bmatrix} ct_4 \\ x_4 \\ y_4 \\ z_4\end{bmatrix} = \begin{bmatrix}\gamma & 0 & \beta\gamma & \\ 0 & 1 & 0 & \\ \gamma & 0 & \gamma & \\ & & & 1\end{bmatrix}\cdot\begin{bmatrix}\gamma & \beta\gamma & 0 & \\ \beta\gamma & \gamma & 0 & \\ 0 & 0 & 1 & \\ & & & 1\end{bmatrix}\cdot\begin{bmatrix}\gamma & 0 & -\beta\gamma & \\ 0 & 1 & 0 & \\ -\beta\gamma & 0 & \gamma & \\ & & & 1\end{bmatrix}\begin{bmatrix}\gamma & -\beta\gamma & 0 & \\ -\beta\gamma & \gamma & 0 & \\ 0 & 0 & 1 & \\ & & & 1\end{bmatrix}\begin{bmatrix}ct\\ x\\ y\\ z\end{bmatrix}$$

七.



对 1→i 块 $\Sigma\vec{F}=0$

$$N_i\cos\theta_i - f_i\sin\theta_i = N_0\cos\theta_0$$

$$N_i\sin\theta_i + f_i\cos\theta_i = +N_0\sin\theta_0 + iG$$

$$N_0\sin\theta_0 = \frac{G}{2}$$

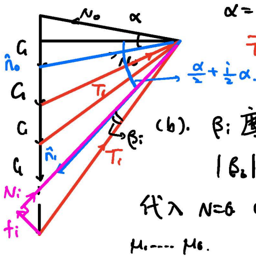
$$\theta_i = \frac{\pi}{2N+1}\left(\frac{1}{2}+i\right)$$

$$\alpha = \frac{\pi}{2N+1}$$

$$\Rightarrow \begin{cases} N_i = N_0\cos\theta_0\cos\theta_i + N_0\sin\theta_0\sin\theta_i + iG\cdot\sin\theta_i \\ \quad = N_0\cdot\cos(\theta_i-\theta_0) + iG\sin\theta_i \\ f_i = N_0\sin\theta_0\cos\theta_i - N_0\cos\theta_0\sin\theta_i + iG\cos\theta_i \\ \quad = -N_0\cdot\sin(\theta_i-\theta_0) + iG\cos\theta_i \end{cases}$$

$$N_0 = \frac{G}{2}\frac{1}{\sin\theta_0}.$$

$$\begin{cases} N_i = \frac{2i+1}{2}G\sin\left(i\alpha+\frac{\alpha}{2}\right) + \frac{G}{2}\cot\left(\frac{\alpha}{2}\right)\cos\left(i\alpha+\frac{\alpha}{2}\right) \\ f_i = \frac{2i+1}{2}G\cos\left(i\alpha+\frac{\alpha}{2}\right) - \frac{G}{2}\cot\left(\frac{\alpha}{2}\right)\sin\left(i\alpha+\frac{\alpha}{2}\right). \end{cases}$$



$$\alpha = \frac{\pi}{2N+1}$$

$$\vec{T}_1 = \vec{N}_1 + \vec{f}_1$$

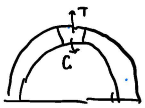
(b). β_i 摩擦角. $|\beta_i|_{max}$. 画图→量角器得

$|\beta_6|$→最大.

代入 $N=6$ $\alpha=\frac{\pi}{13}$. $\mu_i=|\tan\beta_i|$.

μ_1---- μ_6. $\mu_6\to 0.634$. 最大.

c). $N=6$. $\mu=0.65$



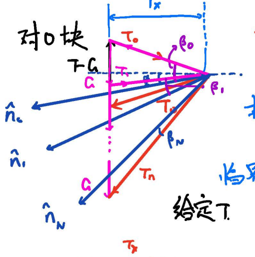
对上 2i+1 块整体→解临界 T_i

取 $\{T_i\}_{min}$

f_i,N_i 之间有关系 T_i 临界时

T_j 可能超临界→辛.

对0块



$$\vec{T}_0 = \vec{N}_0 + \vec{f}_0$$

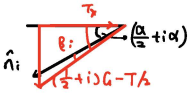
对1块---

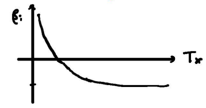
找一个 T_x. 对每个 $|\beta_i|\le\arctan\mu$.

→T 未掉下.

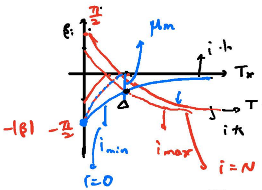
临界点T. T_x 有唯一解.

给定T. $T_x - \beta_i$ $\beta_i = \arctan\frac{(\frac{1}{2}+i)G-T/2}{T_x} - \left(\frac{\alpha}{2}+i\alpha\right)$ =





$T<(2i+1)$



$$\beta_i = \arctan\frac{(\frac{1}{2}+i)G-T/2}{T_x} - \left(\frac{\alpha}{2}+i\alpha\right)$$

$T>(2i+1)G$

$T<(2i+1)G.$

交点位置:

$$\underset{\underset{0.65}{\uparrow}}{\arctan\mu} = \frac{\alpha}{2} + \arctan\frac{T/2-G/2}{T_x} \Leftarrow i=0$$

$$= \left(\frac{1}{2}+N\right)\alpha + \arctan\frac{T/2-(N+\frac{1}{2})G}{T_x} \Leftarrow i=N$$

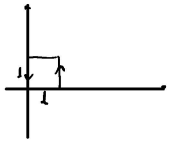
$\Rightarrow T/G = x$. $T_x/G = y$.

$$\begin{cases} x-\frac{1}{2} = \tan\left[\arctan\mu - \tan\frac{\alpha}{2}\right]\cdot y \\ x-\left(6+\frac{1}{2}\right) = \tan\left[\arctan\mu - \tan\left(\frac{1}{2}+6\right)\alpha\right] y \end{cases}$$

$\Rightarrow x=..$

8.

$$B = B_0\cos(\omega t-kx)$$



(1) $$\phi_0 = +l\cdot\int_0^l dx\cdot B_0\cos(\omega t-kx)$$

$$= -\frac{l}{k}B_0\left[\sin(\omega t-kl) - \sin(\omega t)\right]$$

$$\mathcal{E} = -\frac{d\phi_0}{dt}$$

$$= -\frac{l\omega}{k}B_0\left[\cos(\omega t-kl) - \cos\omega t\right]$$

(2). 令匀速度 v.

$$\phi = l\cdot\int_{vt}^{vt+l} dx\cdot B_0\cos(\omega t-kx)$$

$$\phi = -\frac{l}{k}\cdot B_0\left[\sin(\omega t-k(vt+l)) - \sin(\omega t-kvt)\right]$$

$$\mathcal{E} = -\frac{d\phi}{dt} = -\frac{l}{k}B_0[\omega-kv]\cdot\left(\cos((\omega-kv)t-kl) - \cos(\omega-kv)t\right).$$

$$I = \frac{\mathcal{E}}{R}.$$

$$F = I\cdot l\cdot(B_R-B_L)$$

$$F = -\frac{l}{k}B_0^2(\omega-kv)\frac{1}{R}\cdot l\cdot\left(\cos((\omega-kv)t-kl) - \cos((\omega-kv)t)\right)^2$$

$$= mg\mu.$$

$$\left[2\sin\frac{kl}{2}\cdot\sin\left((\omega-kv)t+\frac{kl}{2}\right)\right]^2$$

① $\omega-kv=0$ 后无t. 前 $\omega-kv=0$ $\Rightarrow \Sigma F_\mu=0$ ✗

② $\omega-kv\neq0$ 后有t. 非常数.

⇒ 不可以

$$\sin^2\left((\omega-kv)t+\frac{kl}{2}\right) + \sin^2\left((\omega-kv)t+\frac{kl}{2}+\varphi\right) + \sin^2\left((\omega-kv)t+\frac{kl}{2}+2\varphi\right)$$

$$\Rightarrow \sin^2(\theta) + \sin^2(\theta+\varphi) + \sin^2(\theta+2\varphi)$$

$$= \frac{3}{2} - \frac{1}{2}\left(\underbrace{\cos2\theta + \cos(2\theta+2\varphi) + \cos(2\theta+4\varphi)}_{=0}\right)$$

$$\varphi = \frac{\pi}{3};\ \frac{2\pi}{3}$$